Benchmarking PointNet++ and DGCNN

for 3D Point Cloud Classification

CS5182 Course Project Presentation

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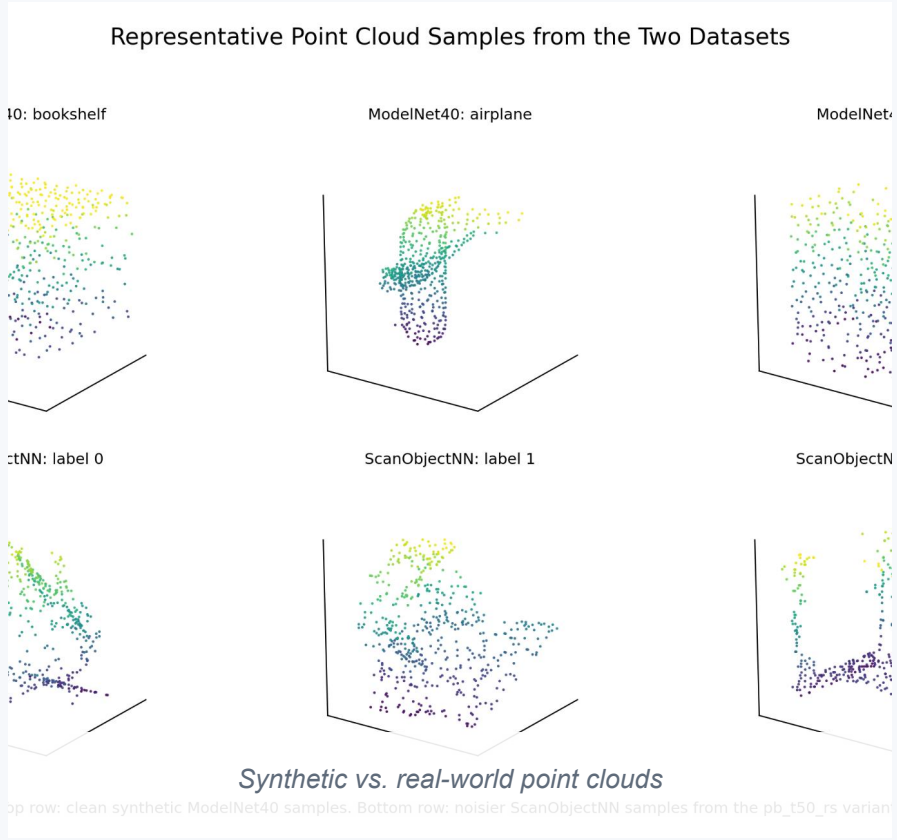
Team of three · group course project

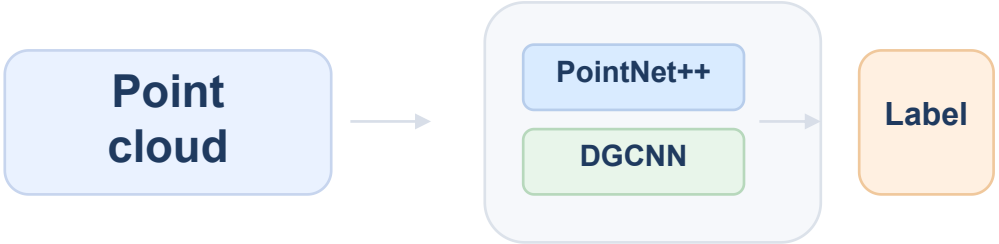
April 2026



Synthetic vs. real-world point clouds

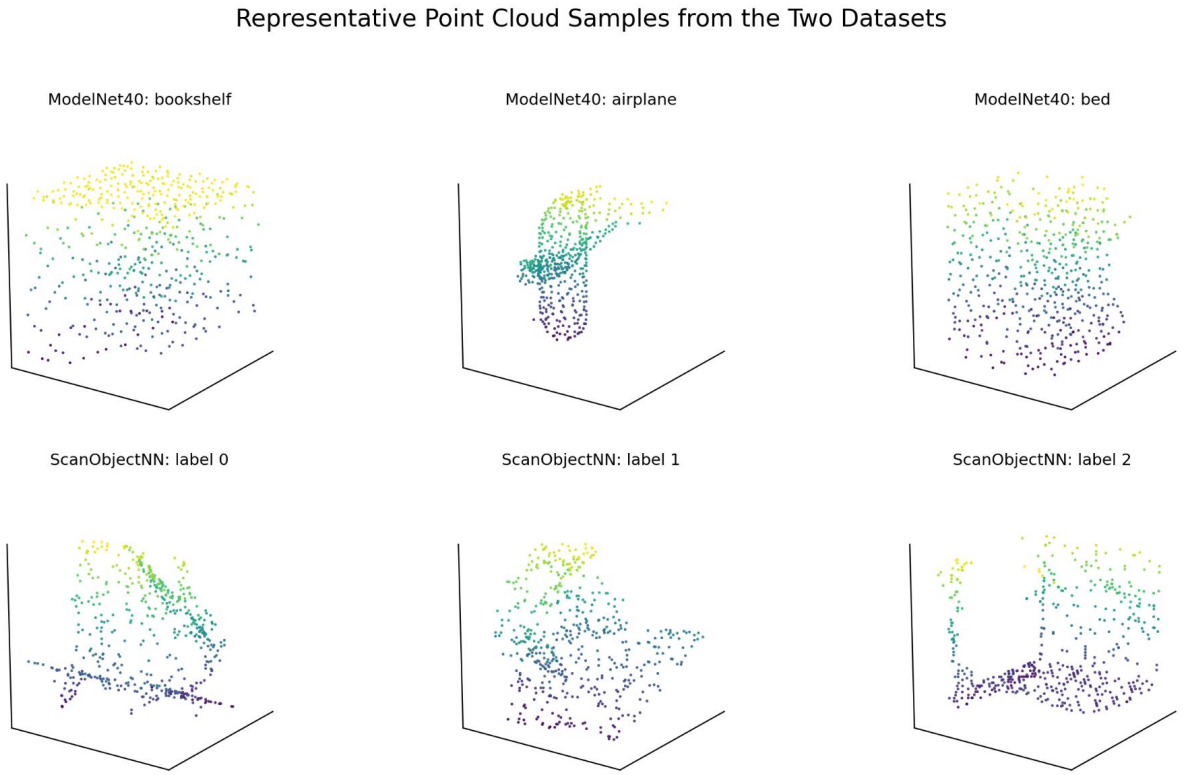
Task, Motivation, and Project Goal

- Task: classify a 3D point cloud into an object category.
- Input: sampled xyz coordinates; Output: class label.
- Compare two representative point cloud classifiers: PointNet++ and DGCNN.
- Evaluate on both a clean benchmark and a harder real-world dataset.
- Goal: build a reproducible local benchmark pipeline, not propose a new method.



Task framing

Datasets



Top row: clean synthetic ModelNet40 samples. Bottom row: noisier ScanObjectNN samples from the pb_t50_rs variant.

- ModelNet40: clean synthetic CAD benchmark with 40 classes.
- ScanObjectNN: real-world scans with clutter, partial observations, and noise.
- ScanObjectNN lightweight is a reduced subset: 4,096 train / 1,024 test.
- This second dataset is visually and conceptually harder than ModelNet40.

Methods: PointNet++ vs DGCNN

PointNet++

Input points → Set abstraction → Classifier

- Hierarchical set abstraction for local feature learning.
- Strong baseline that operates directly on unordered point sets.
- Represents the point-set abstraction family.

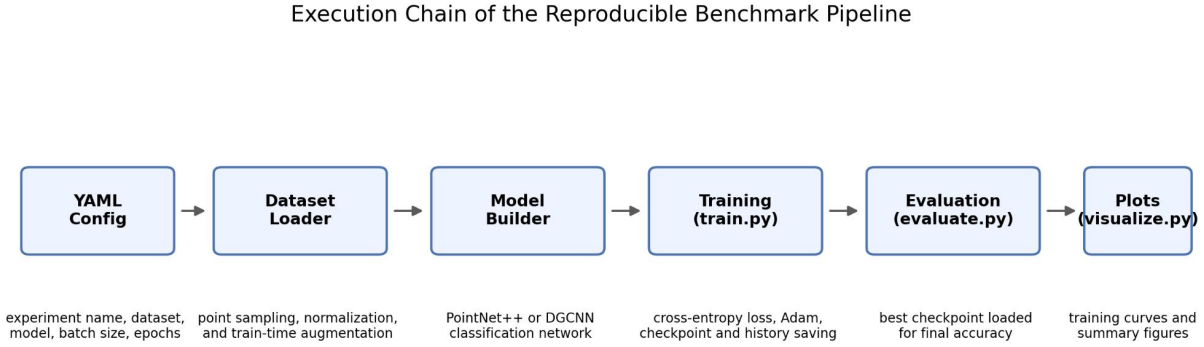
DGCNN

Input points → Dynamic graph → Classifier

- Builds dynamic local graphs and applies EdgeConv layers.
- Models neighborhood relations in feature space.
- Represents the graph-based local geometry family.

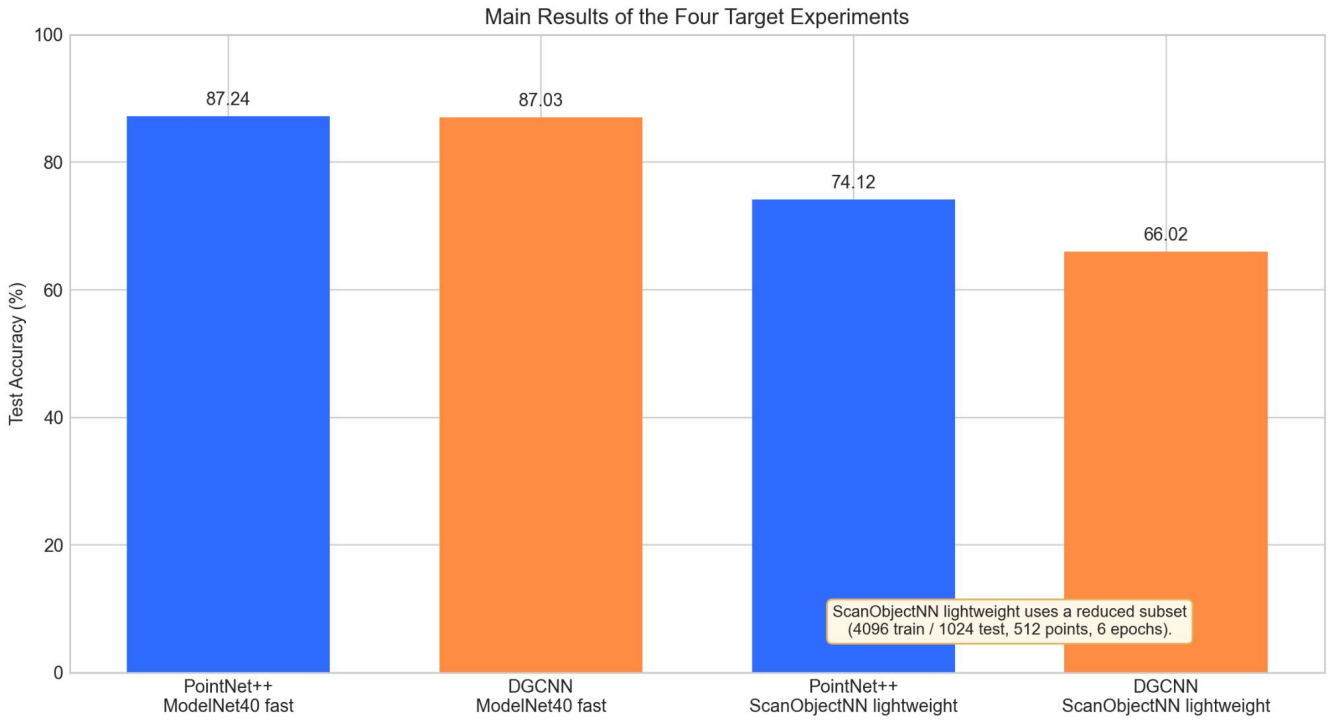
Experimental Setup

- Four unified experiments were used throughout the report and presentation.
- Shared setup: seed 42, Adam, cross-entropy loss, batch size 16, CPU runtime.
- PointNet++ on ModelNet40 fast and DGCNN on ModelNet40 fast use 512 points and 25 epochs.
- PointNet++ on ScanObjectNN lightweight and DGCNN on ScanObjectNN lightweight use 512 points, a reduced subset, and 6 epochs.



Reusable workflow from configuration to training, evaluation, and plots

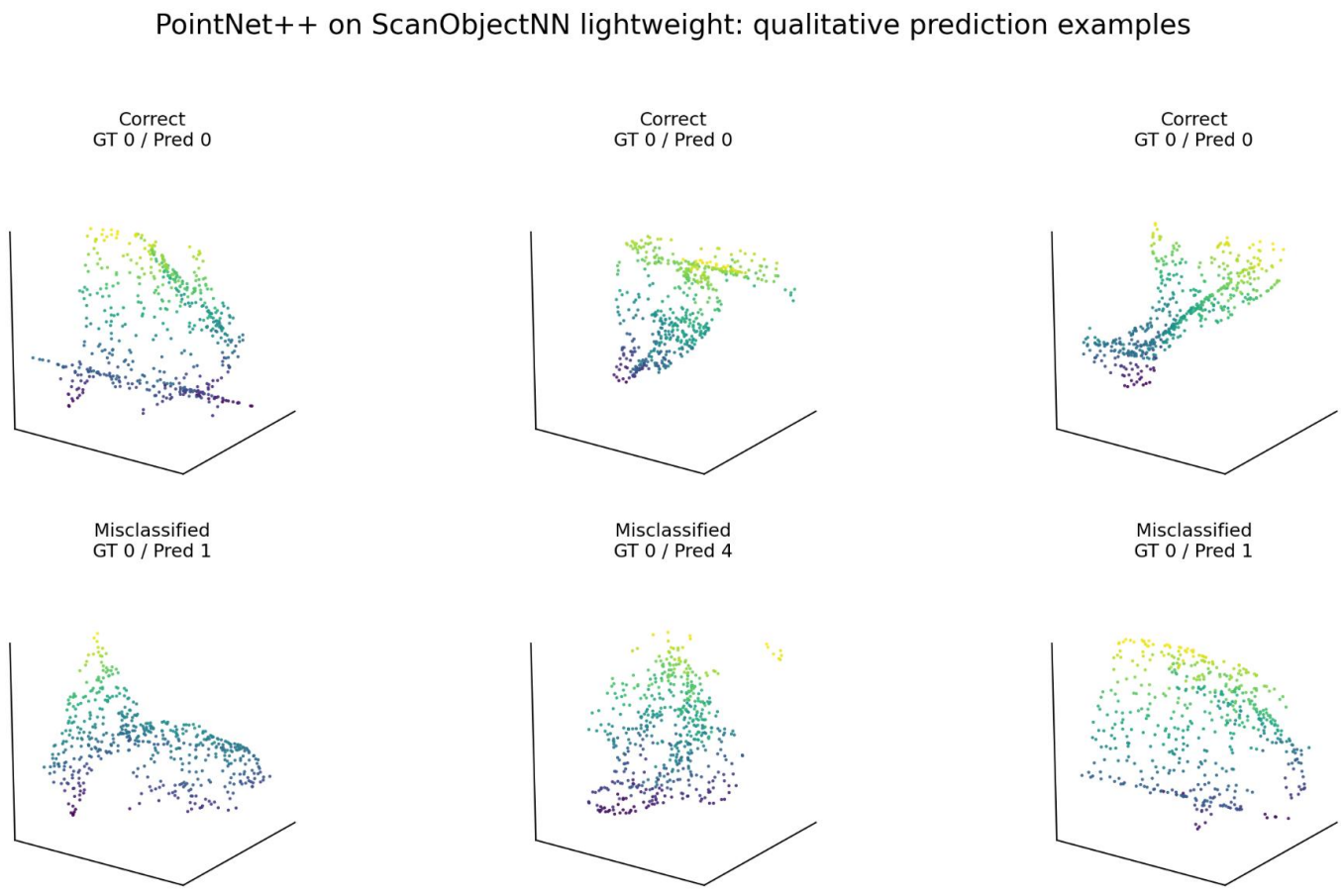
Main Results



- ModelNet40: the two methods are very close (87.24% vs. 87.03%).
- ScanObjectNN lightweight: PointNet++ is higher than DGCNN in our local setting (74.12% vs. 66.02%).
- We do not present this as a universal conclusion on ScanObjectNN in general.
- The second-dataset result mainly supports cross-dataset difficulty and limited-budget analysis.

Interpretation must remain careful because ScanObjectNN lightweight uses a reduced subset and 6 epochs.
This is a limited-budget local comparison, not a full benchmark reproduction.

Qualitative Evidence on ScanObjectNN Lightweight



Top row: correctly classified examples. Bottom row: misclassified examples. GT and Pred denote label indices.

- Top row: correctly classified examples; bottom row: misclassified examples.
- After subsampling to 512 points, some real scans already look incomplete or ambiguous.
- Misclassified cases often show weaker global shape cues and more unstable geometry.
- This supports the claim that ScanObjectNN is harder for local lightweight experiments.

Comparison with Published Results

Experiment	Our Acc.	Paper Acc.	Gap
PointNet++ on ModelNet40 fast	87.24	90.7	-3.46
DGCNN on ModelNet40 fast	87.03	92.9	-5.87
PointNet++ on ScanObjectNN lightweight	74.12	77.9	-3.78
DGCNN on ScanObjectNN lightweight	66.02	78.1	-12.08

- Our local runs are lower than stronger paper numbers, which is expected.
- Reasons: CPU execution, 512 points, reduced subset on ScanObjectNN, and only 6 epochs there.
- So this comparison is informative for trend analysis, not a strict full reproduction.

Key message: honest benchmark comparison rather than overclaiming paper-level reproduction.

Key Takeaways, Limitations, and Teamwork

Takeaways

- We built a reproducible benchmark pipeline for 3D point cloud classification.
- Under matched ModelNet40 fast settings, PointNet++ and DGCNN are close.
- The second dataset is clearly harder and more sensitive to training budget.
- Honest reporting of subset usage is essential for ScanObjectNN lightweight.

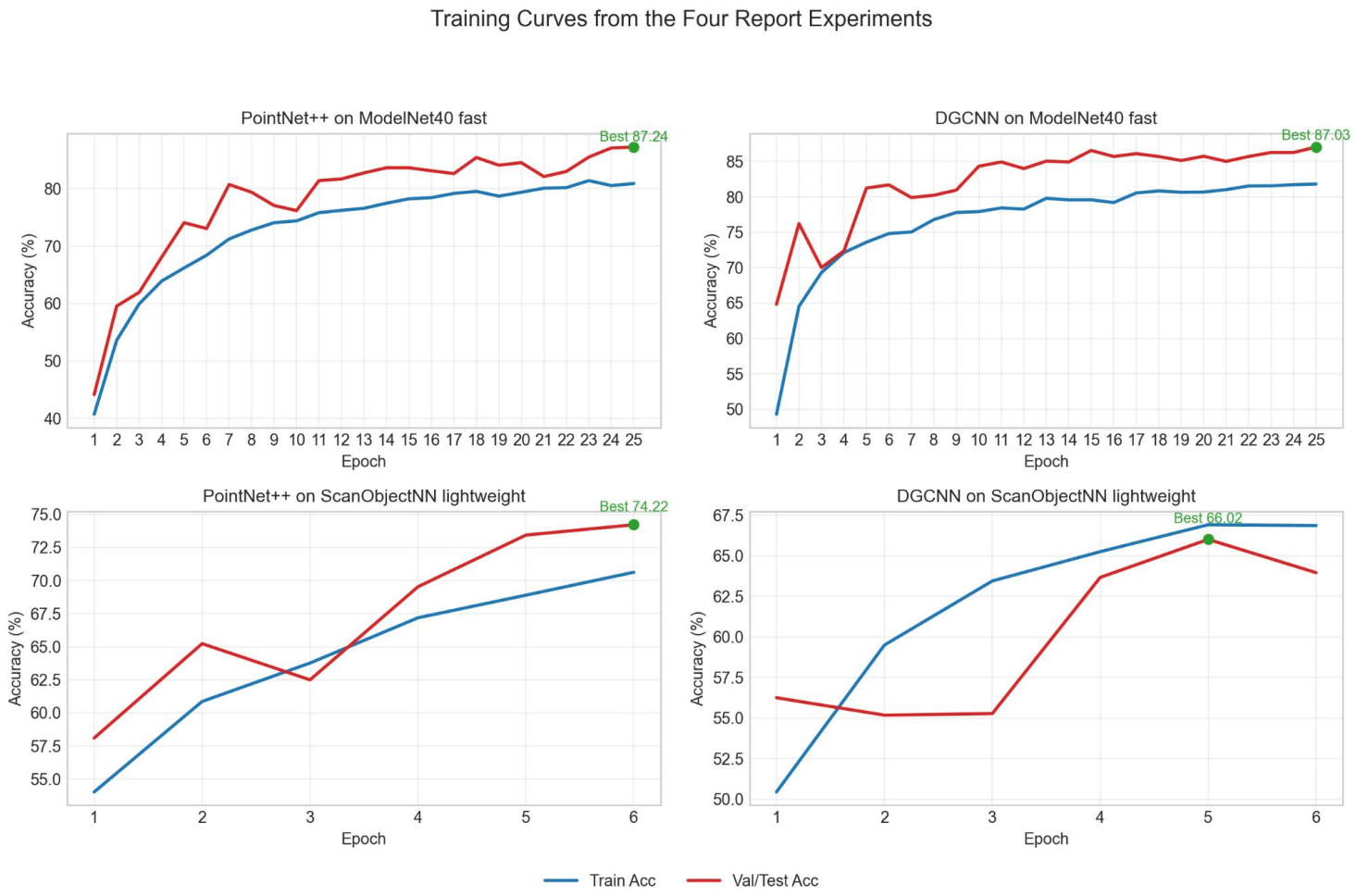
Limitations & teamwork

- Compute limits: CPU only.
- Smaller setup: 512 points.
- ScanObjectNN lightweight uses a reduced subset.
- Short training budget: 6 epochs on ScanObjectNN.
- Three-person project: differentiated ownership, shared integration and presentation.

Future work: full ScanObjectNN runs, GPU training, more points, and stronger error analysis.

Backup: Training Behavior

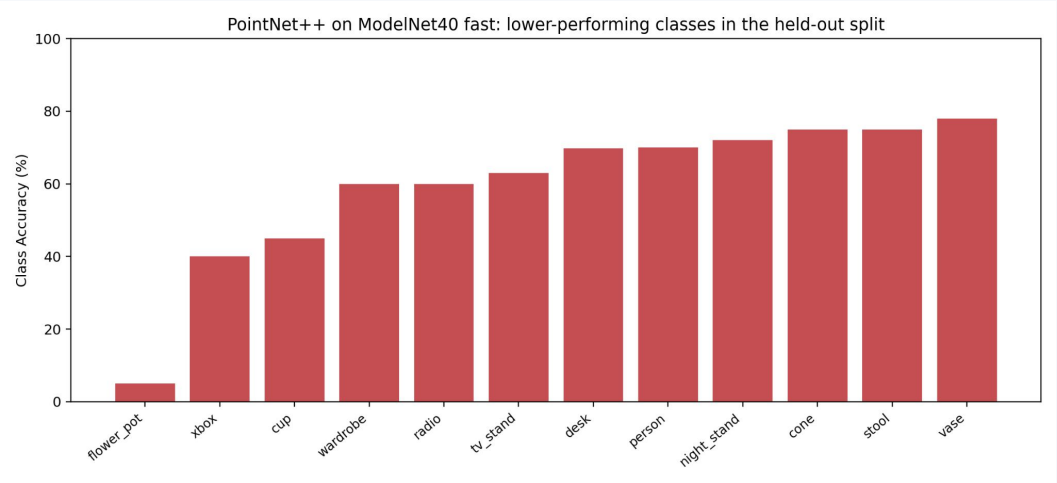
Useful if asked how the final accuracy gap evolved over training.



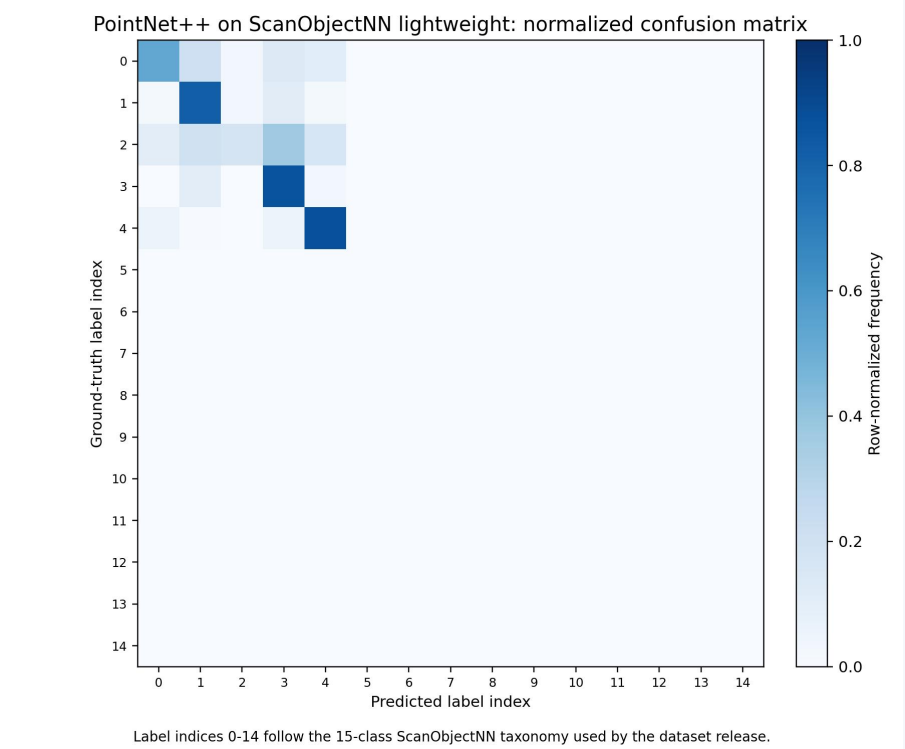
- Both ModelNet40 runs improve steadily across 25 epochs.
- PointNet++ on ScanObjectNN lightweight improves through epoch 6.
- DGCNN peaks earlier at epoch 5 and ends slightly lower.
- This supports the local result gap, but still does not imply a universal ranking.

Backup: Additional Diagnostics

Lower-performing ModelNet40 classes (PointNet++ on ModelNet40 fast)



Normalized confusion matrix (PointNet++ on ScanObjectNN lightweight)



These figures are supplementary and were not part of the core 7-minute presentation line.